

4 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} kx & 0 \leq x < 1, \\ kx^2 & 1 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $k = \frac{6}{17}$. [2]

[illegible]

(b) Find the cumulative distribution function of X . [3]

This image shows a blank sheet of white paper with horizontal ruling lines. There are ten sets of lines across the page. Each set consists of three lines: a solid black line at the top, a dashed black line in the middle, and another solid black line at the bottom. The lines are evenly spaced vertically.



(c) Find the median value of X .

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(d) Find $E\left(\frac{1}{X}\right)$.

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